1. What is digital image? How a digital image can be represented? 3

CT1

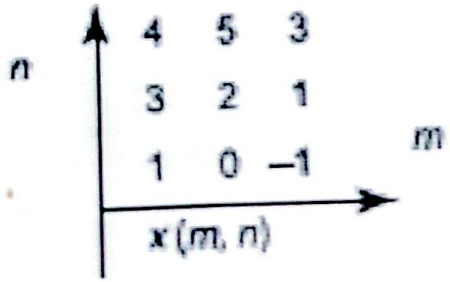
2. Explain the main steps involved in digital image processing 7

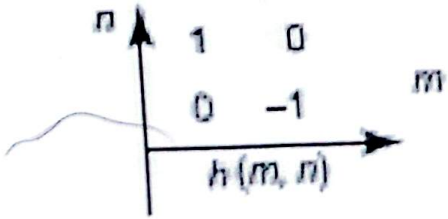
3. Describe the process of image digitization, focusing on sampling and quantization. 5

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1. State the convolution and correlation theorems. 2
2. What is histogram? When will you use local histogram enhancement technique? 3
3. Mention the conditions that should be satisfied for histogram equalization. 2
4. Determine the convolution of $x(m,n)$ with $h(m,n)$ where $x(m,n)$ and $h(m,n)$ are given as follows: 8

C7-3





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CT 4

1.	Define the mean square error (MSE) and signal-to-noise ratio in the context of degraded images.	4
2.	Draw and explain the basic model of the image degradation and restoration process.	6
3.	Explain the six essential noise models in digital image processing with their probability density functions.	10

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1. What is compression ratio? Give the major building blocks of an image compression model. 4
2. Explain the different types of redundancies normally available in an image. 6
3. Construct the Huffman codes for the source symbols the probabilities which are tabulated below. Compute the entropy. 10

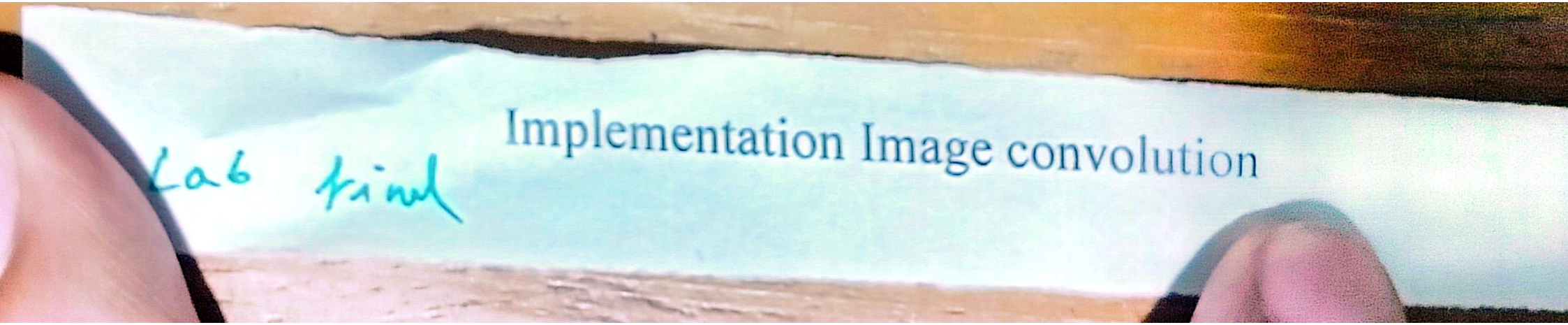
Symbol	S1	S2	S3	S4	S5	S6	S7	S8
Probabilities	0.19	0.21	0.25	0.08	0.16	0.06	0.02	0.03

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Quiz test Course Code: ICT-4105 Course Title: Digital Image Processing Time: 20 minutes

1. Define: Digital image, MSE, Convolution, correlation and pixel. 5
2. What does f(x,y) represent of a digital image? 1
3. In digital image of M rows and N columns and L discrete gray levels, calculate the bits required to store a digi zed image for M=N=32 and L=16 2
4. Explain logical operations (any two) for digital image processing with suitable example. 4
5. Why is image addition used? 2
6. What are the core arithmetic operations? 2
7. What is the kernel size of mean filter? 2
8. What is image enhancement? What are the two main domains of enhancement? 2

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